

SOCIOECONOMIC AND STRUCTURAL FACTORS IN INNOVATION GAPS: EASTERN VS. WESTERN EUROPEAN UNION

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Abstract

To what extent do socioeconomic and structural factors explain the differences in innovation performance, between Eastern and Western European Union countries?”

In this study, using patent applications filed with the European Patent Office (EPO) in 2017 as a proxy for innovation output, we try to investigate how variables such as population, R&D, education, and unemployment rate contribute in explaining the remaining gap between East and West European Union (EU) Member States.

The results show that economic and structural factors affect innovation performance, but they are insufficient to explain the persistent gap between East-West patent applications.

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1. Introduction

As Joseph Schumpeter once pointed out, innovation is above all “a combinatory phenomenon”.

A widely cited definition of innovation comes from the Oslo Manual (OECD, Eurostat, 2018), which defines innovation as “a new or improved product or process, or a combination of both, that differs significantly from previous versions and has been made available to potential users (product) or brought into use by a unit (process).”

Innovation has been crucial in Europe for centuries. The earliest transformative innovations emerged during the late Middle Ages, with the invention, for example, of the printing press by Gutenberg in the 15th century and various advancements in navigation technology that allowed the beginning of the Age of exploration. These earlier innovations not only promoted the beginning of globalization, with the dissemination of knowledge and trade, but also laid ground for all the later industrial and scientific developments.

The Industrial Revolution in the 18th and 19th centuries marked a historical era for European innovation. From the mechanization of textile production to the development of steam engines, and the expansion of railways, societies, economies and labour markets were transformed. Western European countries, in particular, benefited from these earlier industrial advantages, which allowed them to establish a legacy of technological leadership, strong infrastructure, and research institutions.

Eastern Europe, however, followed a somewhat different trajectory. Many of these countries faced delayed industrialization, political disruptions, and economic constraints during the 19th and 20th centuries, which halted the development of innovation ecosystems. The legacies of centrally planned economies, post-war reconstruction, and a later integration into the European Union created structural and socioeconomic conditions that differ significantly from those in the West. These historical factors still contribute nowadays to the innovations gaps observed.

In the modern EU, innovation remains a central policy objective, necessary for competitiveness in the world panorama, economic growth, and social cohesion. Innovation output, in this case, measured through patents, reflects a country's capacity to generate, adopt and diffuse new technologies. Western EU Countries in general maintain stronger innovation systems, while Eastern EU countries, despite recent progress, continue to face structural challenges, brain drain, and smaller innovation networks.

For the purpose of this study, EU Member States are classified into Eastern and Western European countries. Western EU countries include Austria, Belgium, Denmark, Finland, France, Germany, Ireland, Italy, Luxembourg, Netherlands, Portugal, Spain, and Sweden. We will also include Greece, Cyprus, and Malta as part of the Western group due to their long-standing EU membership and economic alignment. The Eastern EU countries comprise Bulgaria, Croatia, Czech Republic, Estonia, Hungary, Latvia, Lithuania, Poland, Romania, Slovakia, and Slovenia.

This project aims to study how certain socioeconomic factors explain the differences in innovation performance, between Eastern and Western EU countries in 2017. By examining innovation performance, through the number of patents, in relation to explanatory factors such as education levels, research and development (R&D), population density, and the unemployment rate, this research expects to identify whether these determinants influence innovation outcomes across regions. Understanding these relationships is important since it enables us to shed light on the innovation disparities that persist, despite efforts to achieve convergence.

2. Literature review

Differences in performance between Western and Eastern EU countries have been a topic of discussion for decades, especially due to their distinct historical background. A large body of literature has exposed these disparities.

Fagerberg (1998), emphasized the critical role of technological opportunities (R&D), human capital accumulation (education), and investment in knowledge-producing sectors in determining a nation's innovation capability. This panorama is closely related with the concept of absorptive capacity (Cohen & Levinthal, 1990), first defined as a firm's "ability to recognize the value of new information, assimilate it, and apply it to commercial ends". Following this idea, a region's ability to recognize, assimilate, and apply new knowledge is highly dependent on its pre-existing stock of human capital and R&D. This suggests that simply increasing R&D might not be sufficient and not yield patent results, unless a proportional increase is made in education, making the efficiency of R&D investment a critical concern.

Building on this, the debate surrounding R&D efficiency is illustrated in the research work of Griliches (1979), who established the concept of diminishing marginal returns to R&D. In transition economies, R&D efficiency may be constrained even if the volume of R&D input increases. The post-socialist shock in some countries led, for example, to the obsolescence of research equipment, which reduces the marginal return to new R&D spending. Consequently, while R&D is an input, its conversion into patent outputs depends heavily on the systematic context, therefore the need to analyze the efficiency differential between East and West.

Radošević (2018), reflects exactly about this enduring East-West divide. He argues that these countries still face structural weaknesses in their science and technology systems, despite deep economic integration with the EU. Radošević highlights persistent issues such as reliance of foreign technology, low R&D intensity, and limited networks between universities and industry. These limitations hinder the development of a self-sustaining ecosystem, which might explain why these countries consistently underperform in innovation scenarios when compared to Western Europe.

Further reinforcing this regional divide, we have the empirical research of Spithoven and Mervlevede (2023), who studied “The productivity impact of R&D and FDI spillovers: characterizing regional path development.”. The idea that R&D investment is not universal but depends heavily on the regional context is reinforced. The results obtained highlight the effectiveness of R&D and the benefits derived from Foreign Direct Investment spillovers can be constructive or not depending on the pre-existing regional structures.

When it comes to education a substantial body of research classifies human capital as a key driver of innovation performance. Diebolt and Hippe (2018) concluded that education levels are strongly associated with higher patenting activity across European countries, in addition to the already accounted for differences caused by economic structure and development. According to their study, regions with a more educated workforce tend to create more knowledge-intensive opportunities, reinforcing the idea that human capital plays a central role in shaping innovation capacity.

Additionally, Pater and Lewandowska (2015) also analysed the relationship between human capital and innovation performance across EU regions, finding that countries with higher educational levels outperform the others in innovation related contexts. Following this idea, the relevance of education when examining variations in innovation outcomes may be crucial.

Research on labour-market conditions also gives us relevant insights in order to comprehend the differences in innovation. Cappelli, Montobbio and Morrison (2021) studied how technological development and the skills of the workforce affect employment outcomes. They concluded that countries with higher levels of innovation tend to have more stable employment, which proves that labour-market performance and innovation might be otherwise connected.

This research aims to use the number of patents as the dependent variable and test the impact of certain socioeconomic factors. Literature provides some justification for the chosen explanatory variables. Research and development is the most direct input into the innovation process. Education levels (human capital) measure the country's absorptive capacity and the skilled labour supply necessary. Unemployment rate is used

as an indicator of the labour market scenario. High unemployment rates may reflect poor utilization of human capital, which might lead to brain drain or reduced firm investment, halting the innovation progress. Population density is used as a proxy for agglomeration economies, where high density facilitates knowledge spillovers critical for innovation. Finally, the inclusion of an East-West Dummy variable is essential for capturing the residual effect of historical legacies and differences in regional path development.

3. Econometrics Model and Data

In this project we used as our main data source the Eurostat database. Eurostat is “the statistical office of the European Union (EU), responsible for providing high-quality, comparable statistics on Europe to support EU policies and enable comparisons between member states and regions”.

The dataset is structured as a cross section for the year of 2017, chosen due to limited data availability for more recent periods. The final sample includes the current 27 EU countries.

The selection of our main independent variable, a regional dummy distinguishing West and Eastern EU countries, was guided by the goal to reflect the central aim of our project, this is to examine whether innovation performance differs between these two groups. Furthermore, control variables that are logically linked to a country’s innovation capacity were included, such as R&D, Education levels, Population Density and Unemployment Rate.

The dependent variable, patent applications to the EPO, and the control variable, population density, were transformed using the logarithm to allow elasticity interpretation, reduce skewness and capture more efficiently non-linear or diminishing marginal effects.

3.1 Dependent variable - Patents

For the dependent variable, we selected the number of patent applications filed under the European Patent Office (EPO) in 2017.

It is important to mention that patents measure invention, rather than innovation in its full definition, which also includes the successful commercialization and diffusion of new products. This being said, not all inventions are commercialized and some innovations do happen without being patented. Nevertheless, patent applications are still commonly used as a proxy for innovation performance.

Moreover, patent applications are typically internationally accepted as a comparable indicator, that is, they can be compared across countries, since they follow the same rules and show inventions that people usually consider valuable enough to protect at a European level. Even though not every patent represents a big technological breakthrough, the applications made to the EPO usually are related with more internationally relevant inventions. That is why they work well as a proxy for comparing innovation performance.

As shown in the descriptive statistics table (see Appendix, figure 1) the average number of EPO patents applications in 2017 was 1822.66, with the minimum and maximum being, respectively, 6.63 for Malta and 18881.7 for Germany. Although the raw patent numbers are mentioned here, in the regression the logarithm of the patent will be used in order to allow a clearer interpretation of the percentage changes.

3.2 Independent Variable

We designated the dummy that distinguishes between Eastern and Western EU Member States as the main independent variable. It takes the value of one if the Member State was previously defined as an Eastern Member State and zero if classified as a Western Member State.

$$EastDummy = \begin{cases} 0 & \text{if West} \\ 1 & \text{if East} \end{cases}$$

Including this variable allows us to test whether there is a systematic difference in the filing of patents between Eastern and Western EU Member States, which is the core objective of our project. We are able to evaluate if belonging to one group instead of another might be associated with higher or lower levels of patent counts, after controlling for other relevant factors.

3.3 Control Variables

Population - log(pop)

This variable represents the total number of inhabitants of each EU Member State. It allows us to capture the size of a country's potential pool of inventors. Larger countries may provide more potent applications, simply because they have a higher

number of people. In order to account for this scale effect, the population variable will be used in its logarithm form.

Research and development - R&D

According to Instituto Nacional de Estatística (INE), R&D is defined as “the set of creative work that is systematically undertaken for the purpose of increasing the volume of knowledge, including the knowledge of man, culture and society, as well as the use of this sum of knowledge to conceive new applications”.

In this study, we use R&D values as a percentage of Gross Domestic Product (GDP), which reflects how much a country's economic output is destined to research activities, and allows the comparison between countries with different economic sizes, since it accounts for relative values instead of absolute levels.

$$R\&D = \frac{R\&D \text{ expenditure}}{GDP} * 100$$

A positive association between R&D and patent applications would be consistent with the idea that countries dedicating a greater amount of their resources to research tend to present higher patent activity.

Education - EDUC

For this variable, we are considering the percentage of people aged 25-64 who have completed at least secondary education.

Presenting the educational level as a percentage of the population allows for an easier comparison among EU countries, allowing the capture of differences in the qualifications of the labour force needed to contribute to innovation activities.

Higher levels of education might improve the conditions under which innovative activity can develop, although the strength of this relationship depends on the whole economic context.

Unemployment rate - UNEMP

For this study we have used the unemployment rate as a share of the labour force aged 15 to 74. This indicator provides a better understanding of the broader market dynamics that may influence innovation activity. High unemployment can reflect economic stress in the labour market, while lower unemployment often happens whenever there are stronger conditions that may support innovation. Although this connection might not be necessarily direct, including this variable will help to account for labour-market differences.

*R&D with Education - R&D*EDUC*

The interaction term between R&D and education allows us to test whether the relationship between R&D and the number of patent applications depends on the education levels of the workforce. This is significant, because the effectiveness of R&D may differ depending on the availability of skilled human capital. In countries, with higher education, R&D might translate more efficiently into patentable innovations, whereas in countries with lower levels, the opposite may happen. With this variable we can study whether education reinforces or conditions the impact of R&D on innovation performance across EU Member States.

4. Violation of assumptions

In multiple linear regression, one should always start by verifying its assumptions in order to check the trustworthiness of the results obtained. This way, as a means to make our interpretation more accurate, we will analyze MLR assumptions one by one, since satisfying them is key for finding BLUE (Best Linear Unbiased Estimator) coefficients, under the Gauss-Markov Theorem.

It is important to remember that we took logs in our model. Applying logarithmic transformation to our variables has likely helped linearize the relationship we are studying and reduce heteroskedasticity. Despite assuming this is happening, we still need to test the MLR assumptions, to ensure the validity of our results.

MLR.1 - Linearity in Parameters

The first assumption requires that the model be linear in the parameters. In our case, this condition is satisfied by construction. Even though our variables involve logarithmic transformations and an interaction term between R&D and education, all coefficients enter the model linearly. Therefore, there is no violation of MLR.1.

In addition, we conducted the Ramsey RESET test to check for possible functional form misspecification such as omitted nonlinearities. Since the test produced a p-value of 0.551 (Appendix, Figure 14), we fail to reject the null hypothesis that the model is correctly specified, meaning there is no evidence of omitted nonlinearities.

MLR.2 - Random Sampling

The data consist of EU Member States and therefore do not come from a random sample but rather from a near-census of the population of interest. In the formal sense, this violates the random sampling. However, because we observe essentially the entire population of interest, the lack of random sampling won't create a critical problem for inference within the EU.

The main limitation is that statistical inference will be less generalizable beyond the context of the EU, and sample variability doesn't behave exactly as it would under true random sampling.

For internal EU comparisons, this lack of randomness won't be a significant concern and the use, later on, of robust standard errors reduces the need to rely on strict sampling assumptions.

MLR.3 - No Perfect Collinearity

According to this assumption, the regressors must not exhibit perfect linear relationships. By looking at our correlation matrix (Appendix, Figure 2), there is no indication of perfect correlation between any pair of variables, meaning the assumption is formally satisfied.

However, the Variance Inflation Factors (VIF) for Model 4 (Appendix, Figure 12) reveal extremely high VIF values for R&D (approximately 213) and the interaction term R&D*Education (approximately 230), as well as a moderately high VIF for Education (close to 9). These do not constitute perfect collinearity, but they clearly indicate severe multicollinearity. This inflates standard errors and reduces statistical significance, although it doesn't bias the coefficient estimates. Given their theoretical relevance, we opted to retain these variables despite the multicollinearity, but we acknowledge that this compromises the precision of the estimates.

MLR.4 - Zero Conditional Mean

The zero conditional mean assumption requires that the error term be uncorrelated with the regressors, which in turn implies that all relevant determinants of the dependent variable must be included in the model. Although we include several key socioeconomic variables, there remain several potentially important factors we cannot control due to data limitations. Examples include institutional quality, the strength of national innovation systems, historical technological legacies and the specific industrial structure of each country.

The fact that the EastDummy remains large and highly significant even after controlling for many variables, suggests that systematic differences between Eastern and Western EU countries are not fully captured. This could reflect true structural divergences, but it may also indicate omitted variable bias. As such, MLR.4 is the most likely assumption to be violated in our study, and this must be considered when interpreting the estimated coefficients.

MLR.5- Homoskedasticity

The homoskedasticity assumption requires that the variance of the error term be constant across observations. To access this, we performed a Breusch–Pagan test (Appendix, Figure 11). The test statistic was 9.79 with a p-value of 0.134, meaning we fail to reject the null of homoskedasticity at the 5% level. Nevertheless, cross-country data are often heteroskedastic by nature (larger countries tend to exhibit larger variance), so some degree of non-constant variance may still be present even if the test doesn't detect it strongly.

For this reason, we re-estimated Model 4 using heteroskedasticity-robust standard errors. The key findings remained unchanged: $\log(\text{population})$ and the East dummy remained statistically significant, indicating that any remaining heteroskedasticity doesn't materially affect our inference. Unemployment however, under robust standard errors, loses its significance- reflecting some limited precision.

MLR.6 - Normality of the Error Term

Normality is not required for OLS estimators to be unbiased, but it is important for valid hypothesis testing in small samples. Because our dataset includes only 27 countries, checking this assumption is relevant. We inspect the distribution of residuals using both a histogram (Appendix, Figure 10) and a Q–Q plot (Appendix, Figure 13). Both show that the residuals approximate a normal distribution without extreme deviations or visible outliers. Even if minor departures from normality exist, given the use of robust standard errors, they are unlikely to seriously affect inference. Therefore, we conclude that the normality assumption appears to be reasonably satisfied for our purposes.

5. Results

We now turn to the empirical results of our analysis. The four econometric models presented below are estimated using Ordinary Least Squares (OLS), which identifies the set of coefficients that minimises the sum of squared residuals between observed and predicted patent output. Each model gradually introduces additional explanatory variables, to investigate how these factors shape cross-country differences in innovation within the EU.

Model 1

$$\log(\text{patent}_i) = \beta_0 + \beta_1 \text{EastDummy}_i + u_i$$

We started our analysis by running the simple linear regression between our two main variables: the dependent variable (y) as the logarithm of patent applications, and a dummy distinguishing between East and West EU Member States as the main independent variable (x), in order to capture the unconditional difference in innovation output between Eastern and Western Europe (Appendix, Figure 3). The estimated coefficient on the EastDummy is about -0.955 , meaning that, *ceteris paribus*, on average, Eastern EU countries have approximately 61.5% fewer patent applications (% difference = $100(e^{-0.955}-1) \approx -61.5\%$). This value is statistically significant at the 1% level (p-value = 0.0089).

The intercept $\beta_0=2.817$ represents the predicted log number of patent applications for a Western EU country. The model has a R^2 of 0.244, meaning that the simple East-West divide explains about 24% in patenting variation.

Model 2

$$\log(\text{patent}_i) = \beta_0 + \beta_1 \log(\text{pop}_i) + \beta_2 \text{EastDummy}_i + u_i$$

In Model 2, we include $\log(\text{pop})$ to examine whether the East–West innovation gap is partly driven by differences in country size (Appendix, Figure 4). The coefficient on $\log(\text{pop})$ is 1.179 and highly significant ($p < 0.001$), indicating that a 1% increase in population is associated with a 1.18% increase in patents. This confirms the applications for patents increase more than proportionally with population, therefore larger countries naturally produce more innovations due to scale effects.

After including population, the coefficient on the *EastDummy* decreases in magnitude to -0.732 in Model 2, but remains highly significant ($p < 0.001$). When converted into a percentage, this coefficient corresponds to about -52% , meaning Eastern EU countries file roughly 52% fewer patents than Western EU countries, even after controlling for population size. Thus, country size explains part of the observed difference, but a substantial East–West innovation gap remains.

The explanatory power of the model increases substantially, with the R^2 increasing to 0.772 in Model 2. The F-statistic (40.55, $p < 0.0001$) confirms that the included variables are jointly relevant predictors.

Model 3

$$\log(\text{patent}_i) = \beta_0 + \beta_1 R\&D_i + \beta_2 Educ_i + \beta_3 \log(\text{pop}_i) + \beta_4 Unempl_i + \beta_5 EastDummy_i + u_i$$

In Model 3, we extend specification by including R&D, education, and unemployment as controls (Appendix, Figure 5). The model provides a strong fit to the data, with an R^2 of 0.95 and an F-statistic of 88.71, $p < 0.001$, indicating that the regressors are jointly highly significant. R&D ($\beta_1 \approx 0.39$, $p < 0.01$) and $\log(\text{pop})$ ($\beta_3 \approx 1.04$, $p < 0.01$) both have strong positive effects on innovation, while unemployment has a negative and significant impact ($\beta_4 \approx -0.033$, $p < 0.05$). Education has a small positive coefficient (0.0079) but is not statistically significant at conventional levels ($p = 0.18$), suggesting that education alone does not directly predict patenting when controlling for the other variables. This will support our idea that education may influence innovation primarily through interaction with R&D, which will be tested in the following model.

Importantly, the East–West dummy remains negative and significant ($\beta_6 \approx -0.60$, $p < 0.01$), implying that, even after controlling for R&D, education, population and unemployment, Eastern EU countries still file around 45% fewer patents than Western countries ($100(e^{-0.599} - 1) \approx -45\%$). This suggests that structural differences between East and West persist beyond observable inputs.

Model 4

$$\begin{aligned}\log(\text{patent}_i) = & \beta_0 + \beta_1 R\&D_i + \beta_2 Educ_i + \beta_3 (R\&D_i \times Educ_i) \\ & + \beta_4 \log(\text{pop}_i) + \beta_5 Unempl_i + \beta_6 EastDummy_i + u_i\end{aligned}$$

In Model 4, we estimate the full specification by adding the interaction between R&D expenditure and education. This allows the effect of R&D on innovation to vary depending on a country's education level (Appendix, Figure 6). Because of the interaction term, the marginal effect of R&D on log patents is obtained by taking the partial derivative of $\log(\text{patents})$ with respect to R&D: Marginal Effect of R&D = $\beta_1 + \beta_3 \times \text{Education}$.

At the sample mean of education ($\approx 80\%$) (Appendix, Figure 3), this gives: ME R&D = $1.0536 - 0.0080 \times 80 \approx 0.41$. So, for a country with average education, a one-unit increase in R&D is associated with about a 0.41 increase in log patent output. Because β_3 is negative, the marginal effect is larger in lower-education countries (e.g., 0.57 when education = 60) and smaller in high-education countries (e.g., 0.33 when education = 90).

Although the coefficients on R&D, education, and the interaction are not statistically significant (p-values > 0.10), this is not unexpected given the small sample (27 countries) and the number of regressors, thus it should be interpreted with caution.

According to the other variables, $\log(\text{pop})$ remains highly significant ($p < 0.001$), confirming that larger countries produce more patents; unemployment is significantly negative, indicating that weak labor market conditions correlate with lower innovation output and EastDummy remains negative and statistically significant ($p < 0.01$), meaning that even after controlling for R&D, education, population and unemployment, Eastern EU countries still file significantly fewer patents. The remaining gap is around -45% .

Model 4 achieves an R^2 of 0.956 and an Adjusted R^2 of 0.947, meaning the full specification explains almost all of the cross-country variation in innovation output. The F-statistic is highly significant, confirming that the regressors are jointly relevant.

Robustness

To ensure the reliability of our regression results and verify whether the OLS assumptions are satisfied, we perform several robustness checks and diagnostic tests.

To assess heteroskedasticity, we estimate a Breusch–Pagan test (Appendix, Figure 11). We have a BP=9.792, for 6 degrees of freedom and a p-value=0.1337. The statistic is below the 5% critical chi-square value ($\chi^2_{0.95,6} = 12.59$), so we fail to reject the null hypothesis of homoskedasticity. This suggests that heteroskedasticity is not a serious concern in our model. Although the Breusch–Pagan test does not indicate strong heteroskedasticity, we report heteroskedasticity-robust standard errors as a precaution, in line with standard practice in cross-country regressions.

To check the sensitivity of our results to potential heteroskedasticity, we re-estimated Model 4 using heteroskedasticity-robust standard errors (Appendix, Figure 8 and 9). Because the sample is small ($N = 27$) and the dataset includes some clear high-leverage countries (e.g. Germany vs Malta), we used HC3 rather than HC1 in RStudio. The robust standard errors differ somewhat from the classical OLS ones, for example the intercept (0.727 vs. 0.928), R&D (0.884 vs. 0.740), and the EastDummy (0.252 vs. 0.166), which is expected when moving towards a more conservative estimator.

After using Robust SE, the unemployment variable is no longer statistically significant ($p = 0.11$), reflecting the limited precision that already existed in the OLS estimates. The variables R&D, Education and the interaction between are still not significant like before. Importantly, the coefficients themselves do not change, only the standard errors and the associated test statistics are affected, so the meaning of the model remains the same.

Overall, the robustness check shows the strong effect of population size and the persistent East-West innovation gap, which remains stable when using robust standard errors. This suggests that our key findings are not driven by heteroskedasticity and remain reliable even under a more conservative inference approach.

To assess potential multicollinearity issues in Model 4, we computed Variance Inflation Factors (VIF), (Appendix, Figure 12). The results indicate that R&D (VIF ≈ 213) and the R&D $\times$ Education interaction term (VIF ≈ 230) exhibit extremely high VIF values, far above the common thresholds of 5 or 10. This confirms that the interaction term is strongly correlated with its component variables, which is expected when including a product term in the regression. Education shows a moderate VIF (≈ 9), while the remaining regressors: log(pop) (≈ 1.47), unemployment (≈ 1.53), and the East dummy (≈ 3.40), fall well within acceptable limits, indicating no multicollinearity concerns for those variables. Overall, MLR.3 is not violated (no perfect multicollinearity), although the very high VIFs for R&D and the interaction term indicate strong but not perfect collinearity, which inflates standard errors, suggesting they need to be interpreted with caution.

Residual diagnostics indicate no major violations of normality. A Q-Q plot shows the residuals closely following the normal reference line (Appendix, Figure 13).

We also conducted the Ramsey RESET test to check whether Model 4 suffers from functional form misspecification, such as omitted nonlinear terms or an incorrect functional relationship (Appendix, Figure 14). The test produces a statistic of 0.616 and a p-value of 0.551. Since the p-value is well above conventional significance levels, we do not reject the null hypothesis that the model is correctly specified. This suggests that there is no evidence of functional form errors, and the linear specification used in Model 4 appears appropriate.

6. Conclusion

As previously mentioned, the aim of this research is to study how certain socioeconomic factors can explain the differences between Eastern and Western EU countries in terms of innovation in 2017. Based on existing studies and our hypothesis, we expected that Western EU countries would have higher levels of innovation compared to Eastern EU countries. Moreover, we also expected that the control variables would help us understand these differences, assuming that all variables, except for the unemployment rate, which we expected to have a negative relationship, would have a positive impact on patents.

In our first model, we confirmed that, as predicted, Eastern EU countries have lower levels of patent applications.

Models 2 and 3, which included our control variables confirmed the structural differences between these groups of countries while improving our model, as the inclusion of these variables allowed us to better understand these differences. Population size was confirmed to have a significant positive impact on innovation, with larger countries filing more patent applications. Additionally, the unemployment rate showed a consistent negative impact on our dependent variable, as expected.

After including our interaction variable, we noticed that the effect of R&D on patent applications might not depend on the level of education, nor on these variables individually, as we initially predicted. However, we cannot be certain of this due to our sample size and the number of regressors.

From an overall perspective, we were able to find that the innovation performance of the EU is indeed influenced by socioeconomic factors. Legislators should focus on reducing the innovation gap between Eastern and Western EU countries by strengthening national innovation systems and improving institutional quality. These structural improvements could help increase innovation performance and reduce these disparities. Future research could use panel data to track innovation over time and examine additional factors and interactions that contribute to these differences.

7. Appendix

Column1	min	max	mean	sd	variance
log patent	0,821513528	4,276041093	2,428396097	0,968228449	0,937466329
education	48	94,8	79,97777778	12,12795034	147,0871795
logpopulation	5,662167356	7,916567919	6,861733762	0,60397671	0,364787866
rd	0,51	3,4	1,568148148	0,889714894	0,791592593
unemployment	2,9	21,8	7,874074074	4,075970162	16,61353276
patent	6,63	18881,7	1822,66	3964,15	15714463,3

Figure 1 - Overall Descriptive Statistics (2017)

Correlation Matrix	Education	R&D	Unemployment rate	log(population)	log(patent)
Education	1	0,165973069	-0,352438231	-0,07583832	-0,014359317
R&D	0,1659731	1	-0,220954467	0,359471714	0,78567137
Unemployment rate	-0,352438	-0,220954467	1	0,166046261	-0,060554219
log(population)	-0,075838	0,359471714	0,166046261	1	0,794951363
log(patent)	-0,014359	0,78567137	-0,060554219	0,794951363	1

Figure 2 - Correlation Matrix

SUMMARY OUTPUT								
Regression Statistics								
Multiple R	0,493669516							
R Square	0,243709591							
Adjusted R Square	0,213457975							
Standard Error	0,858694745							
Observations	27							
ANOVA								
	df	SS	MS	F	Significance F			
Regression	1	5,940207937	5,940207937	8,056084962	0,008872818			
Residual	25	18,43391661	0,737356665					
Total	26	24,37412455						
	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95,0%	Upper 95,0%
Intercept	2,817311634	0,214673686	13,12369338	1,02992E-12	2,375182901	3,259440367	2,375182901	3,259440367
EastDummy	-0,954610863	0,336329029	-2,838324323	0,008872818	-1,647293464	-0,261928262	-1,647293464	-0,261928262

Figure 3 - Summary Output (Model 1)

SUMMARY OUTPUT								
Regression Statistics								
Multiple R	0,878442123							
R Square	0,771660564							
Adjusted R Square	0,752632277							
Standard Error	0,481558834							
Observations	27							
ANOVA								
	df	SS	MS	F	Significance F			
Regression	2	18,80855069	9,404275343	40,55333982	2,00896E-08			
Residual	24	5,565573865	0,231898911					
Total	26	24,37412455						
	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95,0%	Upper 95,0%
Intercept	-5,365922856	1,105109977	-4,855555525	5,99673E-05	-7,646757748	-3,085087963	-7,646757748	-3,085087963
log(pop)	1,179361627	0,158319729	7,449239806	1,08918E-07	0,852605766	1,506117488	0,852605766	1,506117488
EastDummy	-0,731814238	0,190971013	-3,83206973	0,000804551	-1,125959037	-0,337669439	-1,125959037	-0,337669439

Figure 4 - Summary Output (Model 2)

SUMMARY OUTPUT								
Regression Statistics								
Multiple R	0,977135225							
R Square	0,954793249							
Adjusted R Square	0,944029736							
Standard Error	0,229063829							
Observations	27							
ANOVA								
	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>			
Regression	5	23,27224956	4,654449912	88,70647685	2,15453E-13			
Residual	21	1,101874989	0,052470238					
Total	26	24,37412455						
	<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>	<i>Lower 95%</i>	<i>Upper 95%</i>	<i>Lower 95,0%</i>	<i>Upper 95,0%</i>
Intercept	-5,444600514	0,706826707	-7,702878883	1,50257E-07	-6,914527119	-3,974673909	-6,914527119	-3,974673909
R&D	0,386521549	0,086927205	4,446496895	0,000223416	0,205746529	0,567296569	0,205746529	0,567296569
Educ	0,00786069	0,005704058	1,378087376	0,182684106	-0,004001548	0,019722928	-0,004001548	0,019722928
log(pop)	1,040335182	0,08658779	12,01480227	7,11793E-11	0,860266014	1,22040435	0,860266014	1,22040435
Unempl	-0,032536841	0,013237693	-2,457893611	0,022750813	-0,060066131	-0,005007551	-0,060066131	-0,005007551
EastDummy	-0,599185284	0,165438813	-3,621793898	0,001598856	-0,943234131	-0,255136438	-0,943234131	-0,255136438

Figure 5 - Summary Output Model 3

```
Call:
lm(formula = logpatent ~ RD + Education + RD_Education + logpopulation +
    Unemployment + EastDummy, data = df)

Residuals:
    Min       1Q   Median       3Q      Max
-0.34869 -0.15748 -0.00828  0.16210  0.34207

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  -5.987360   0.928201  -6.450 2.73e-06 ***
RD             1.053561   0.740212   1.423  0.1701
Education     0.016529   0.011138   1.484  0.1534
RD_Education  -0.007987   0.008802  -0.907  0.3750
logpopulation  1.017521   0.090515  11.241 4.27e-10 ***
Unemployment  -0.035497   0.013688  -2.593  0.0174 *
EastDummy     -0.601982   0.166167  -3.623  0.0017 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.23 on 20 degrees of freedom
Multiple R-squared:  0.9566,    Adjusted R-squared:  0.9436
F-statistic: 73.44 on 6 and 20 DF, p-value: 1.45e-12
```

Figure 6 - Summary Output (Model 4)

```

studentized Breusch-Pagan test

data:  model4
BP = 9.7921, df = 6, p-value = 0.1337

```

Figure 7 - Breusch-Pagan Test for Heteroskedacity

t test of coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	-5.9873603	0.7268626	-8.2373	7.399e-08	***
RD	1.0535611	0.8837984	1.1921	0.24718	
Education	0.0165292	0.0107704	1.5347	0.14053	
RD_Education	-0.0079875	0.0103793	-0.7696	0.45055	
logpopulation	1.0175205	0.1125958	9.0369	1.688e-08	***
Unemployment	-0.0354975	0.0211831	-1.6757	0.10935	
EastDummy	-0.6019824	0.2518425	-2.3903	0.02679	*

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Figure 8 - HC3 Robust SE t-tests and p-values (Model 4)

	OLS_SE	Robust_SE
(Intercept)	0.928201	0.726863
RD	0.740212	0.883798
Education	0.011138	0.010770
RD_Education	0.008802	0.010379
logpopulation	0.090515	0.112596
Unemployment	0.013688	0.021183
EastDummy	0.166167	0.251843

Figure 9 - OLS SE vs Robust SE comparison (Model 4)

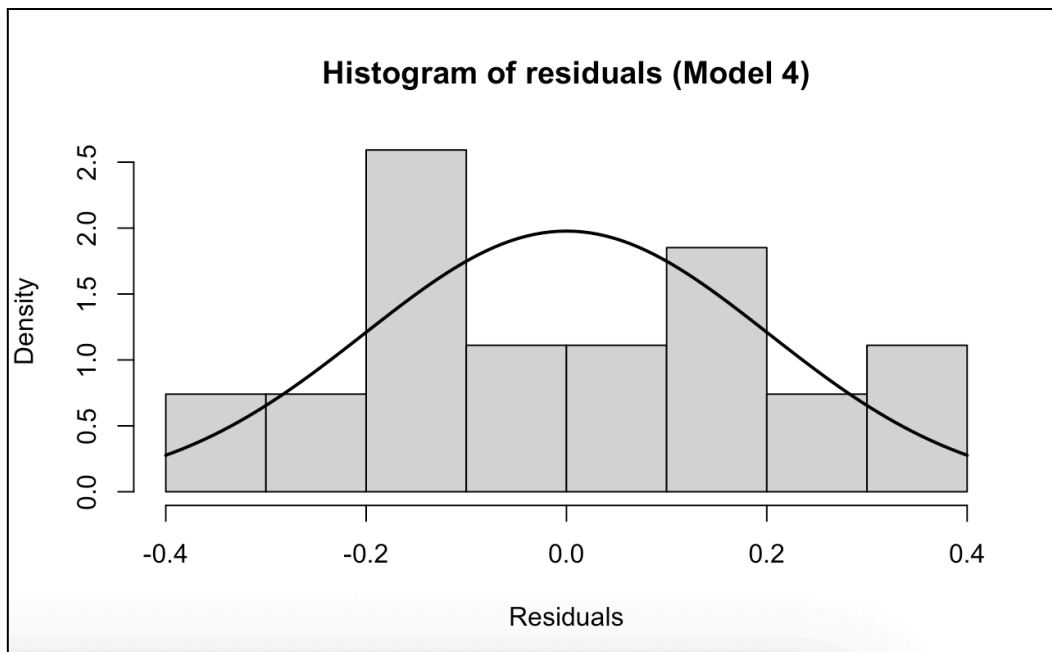


Figure 10 - Histogram of Residuals with Normal Curve

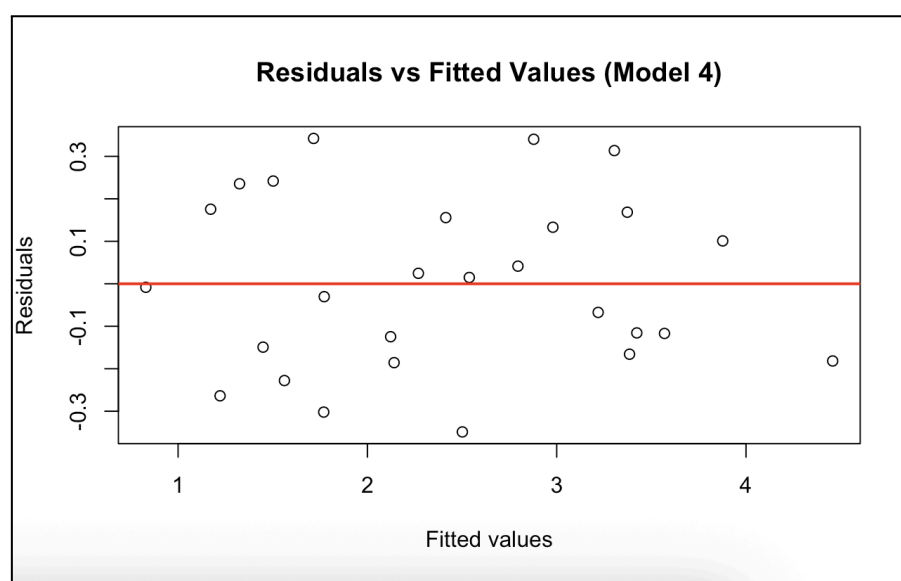


Figure 11 - Residuals vs Fitted Values (Model 4)

> vif(model4)					
RD	Education	RD_Education	logpopulation	Unemployment	EastDummy
213.112641	8.965886	229.756025	1.468523	1.529491	3.401419

Figure 12 - VIF Test for Multicollinearity

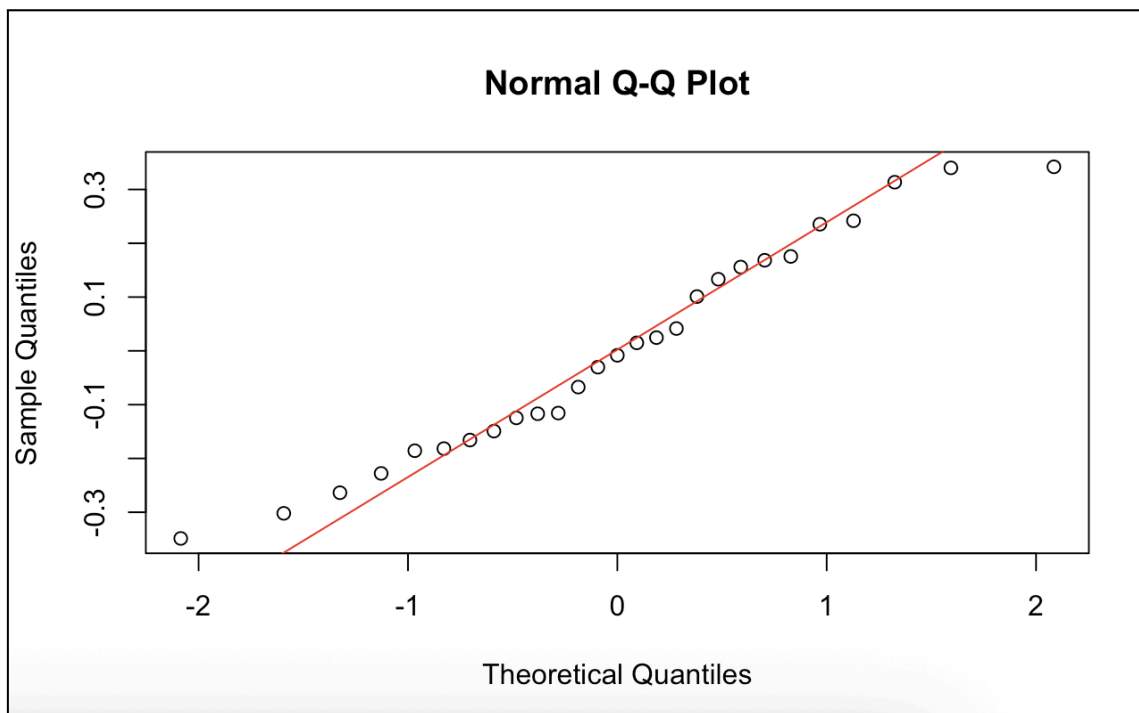


Figure 13 - Normal Q-Q Plot for visual normality check (Model 4)

RESET test

data: model4
RESET = 0.61604, df1 = 2, df2 = 18,
p-value = 0.5511

Figure 14 - RESET test to check whether Model 4 is correctly specified

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